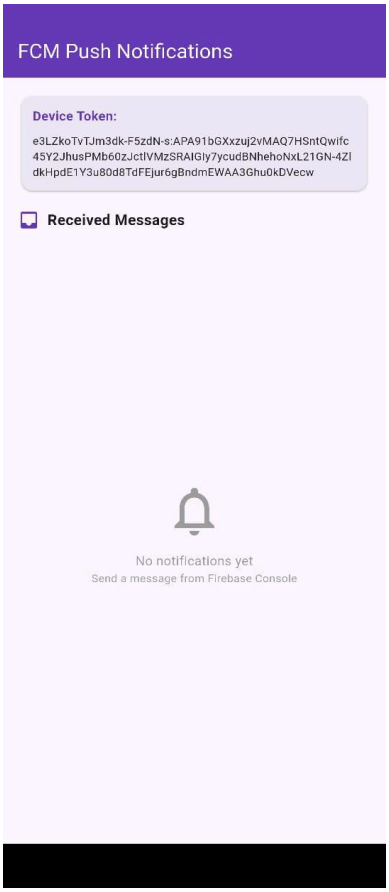
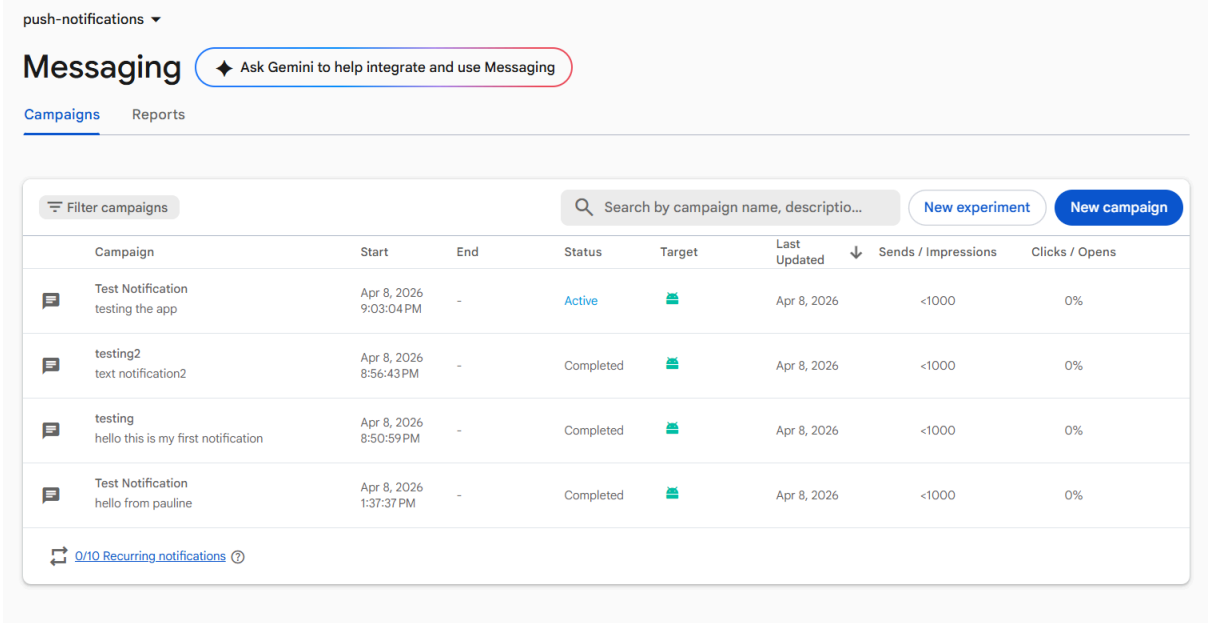
1.App Home Screen (with Device Token)



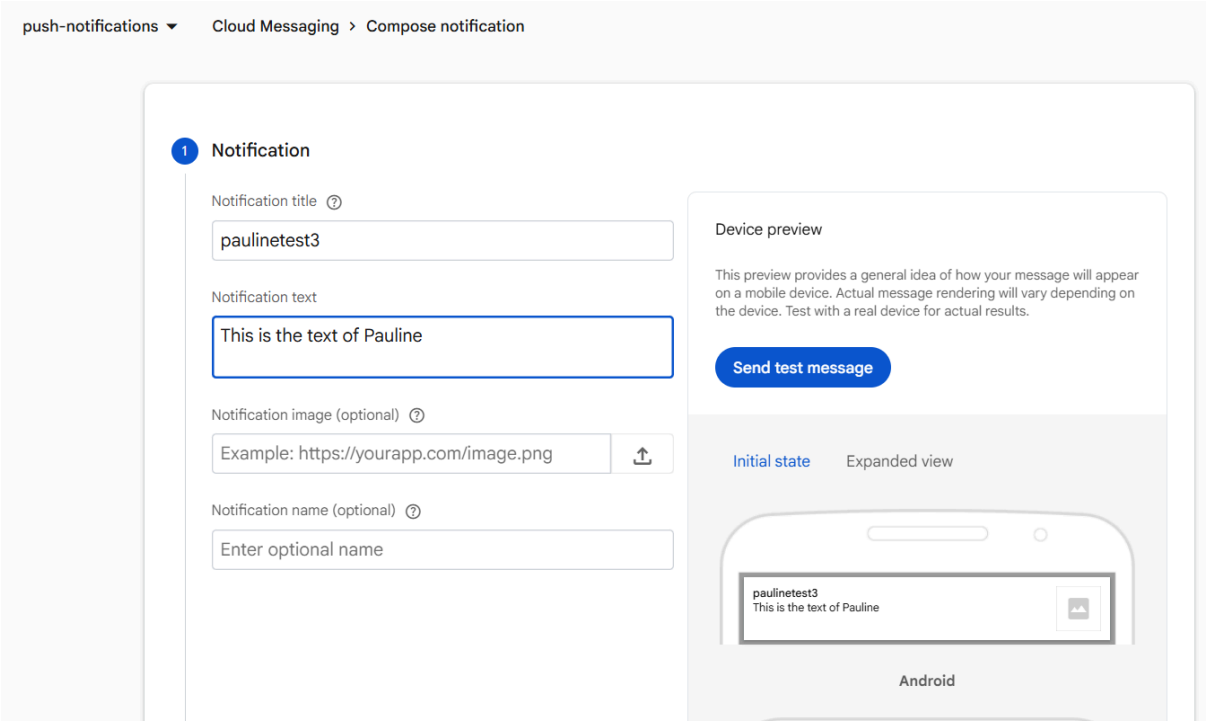
App home screen displaying the device token for FCM testing.

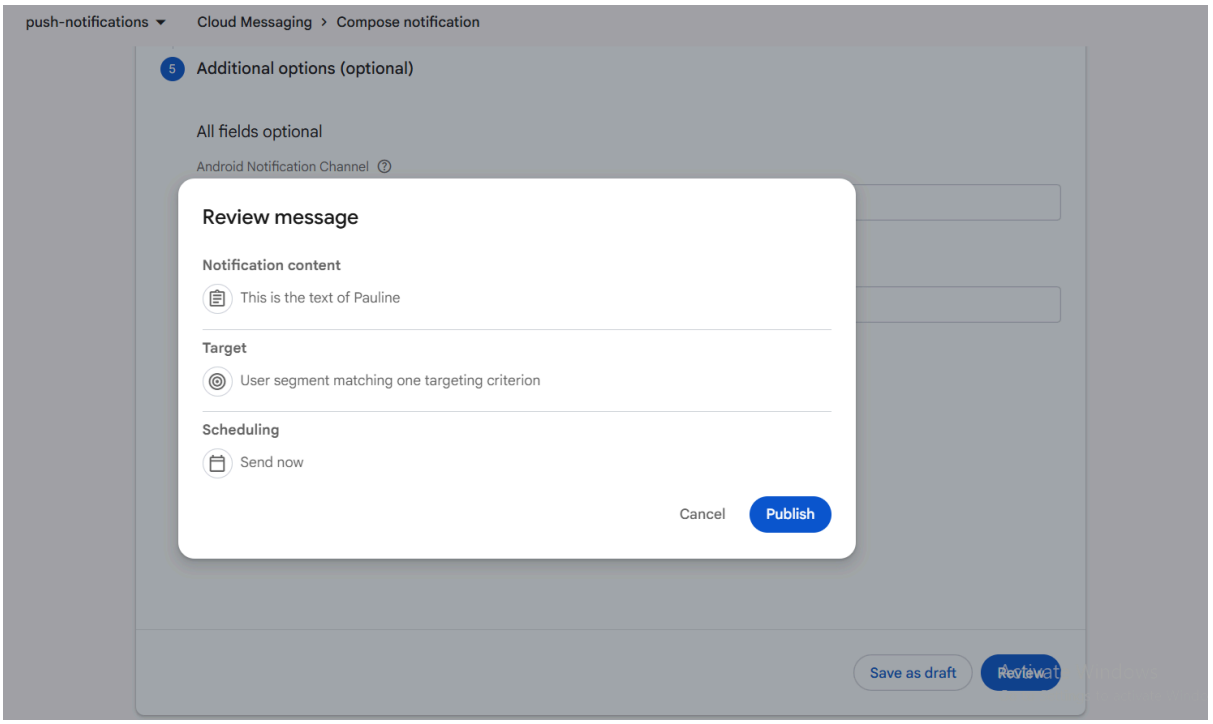
2. Firebase Console – Cloud Messaging Dashboard



Firebase Console: Cloud Messaging dashboard for sending push notifications.through creating new campaign.

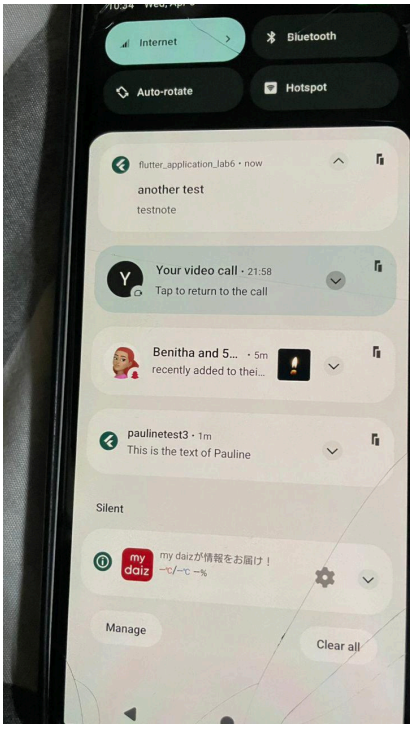
3.Firebase Console – Create New Message





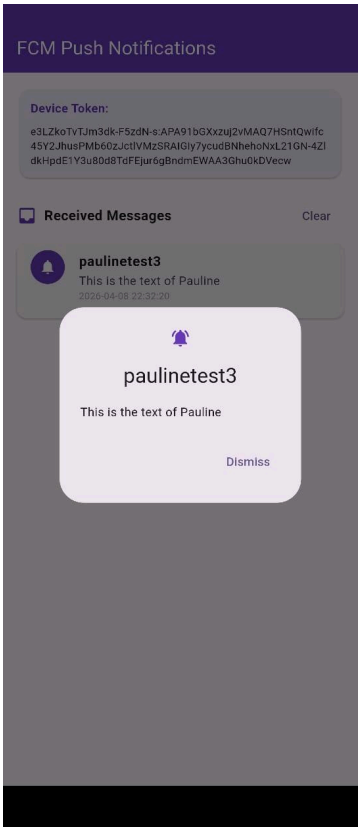
Firebase Console: Creating a new push notification message targeted to the device.

4.Received notification



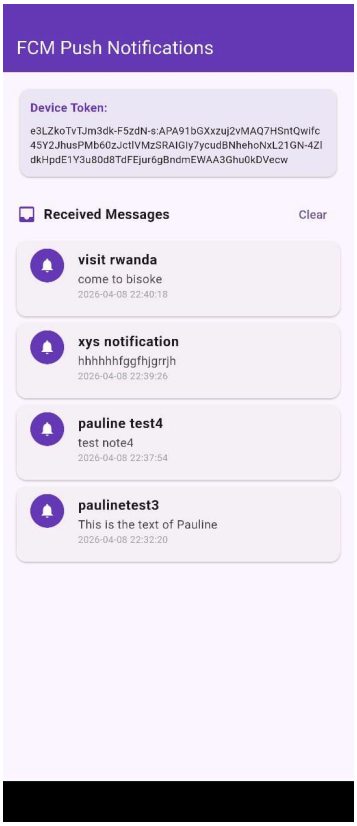
The notification came to my phone

5. App Receiving Notification (Popup)



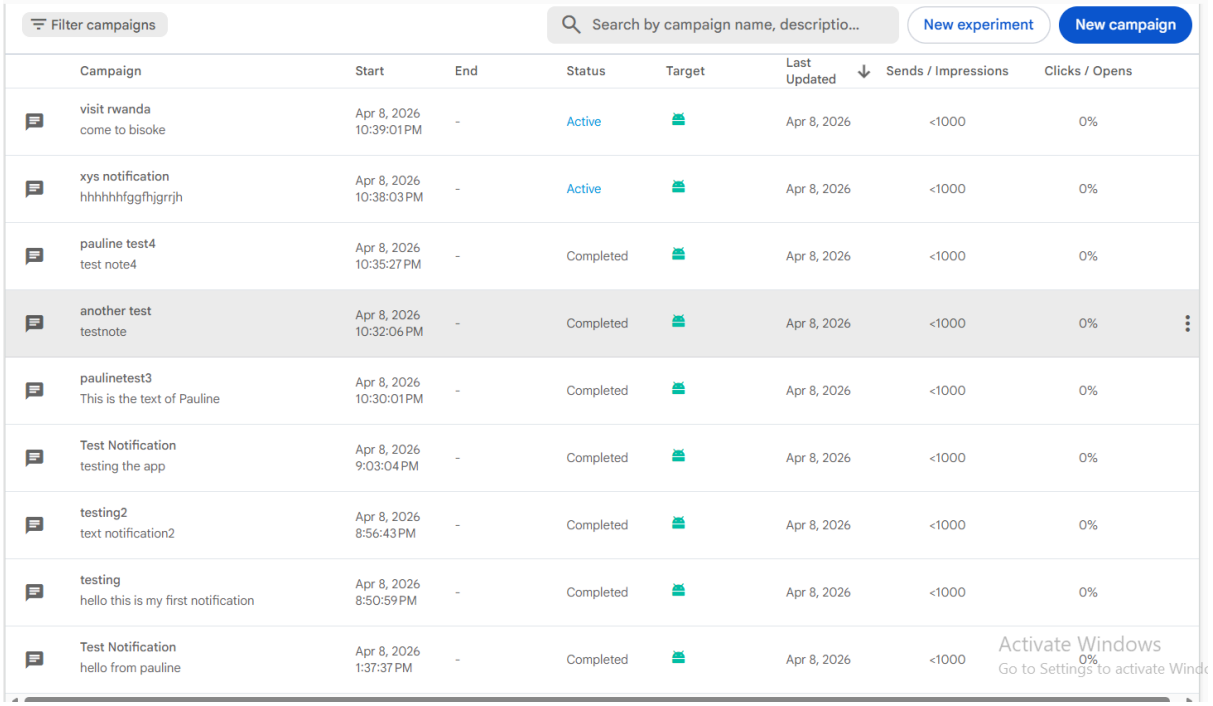
Popup dialog shown when a push notification is received while the app is open.

6. App Notification List



List of received notifications displayed in the app UI (persisted with Hive).

7. Firebase Console – Message History



Filter campaigns | Search by campaign name, descriptio... | New experiment | New campaign

Campaign	Start	End	Status	Target	Last Updated	Sends / Impressions	Clicks / Opens
visit rwanda come to bisoke	Apr 8, 2026 10:39:01 PM	-	Active		Apr 8, 2026	<1000	0%
xys notification hhhhhhfggfhjgrrjh	Apr 8, 2026 10:38:03 PM	-	Active		Apr 8, 2026	<1000	0%
pauline test4 test note4	Apr 8, 2026 10:35:27 PM	-	Completed		Apr 8, 2026	<1000	0%
another test testnote	Apr 8, 2026 10:32:06 PM	-	Completed		Apr 8, 2026	<1000	0%
paulinetest3 This is the text of Pauline	Apr 8, 2026 10:30:01 PM	-	Completed		Apr 8, 2026	<1000	0%
Test Notification testing the app	Apr 8, 2026 9:03:04 PM	-	Completed		Apr 8, 2026	<1000	0%
testing2 text notification2	Apr 8, 2026 8:56:43 PM	-	Completed		Apr 8, 2026	<1000	0%
testing hello this is my first notification	Apr 8, 2026 8:50:59 PM	-	Completed		Apr 8, 2026	<1000	0%
Test Notification hello from pauline	Apr 8, 2026 1:37:37 PM	-	Completed		Apr 8, 2026	<1000	0%

Activate Windows
Go to Settings to activate Windo

Firebase Console: History of sent push notification campaigns

Marie Pauline Ishimwe

COE 2026 flutter project Cloud Messaging

Observation

- I learned to integrate firebase cloud Messaging in flutter app to receive notifications
- I discovered importance of handling notification permission, device token and different app state (foreground & background , terminated)
- using hive for local storage allowed me to persist notification remained visible even after restart of app
- The most challenging was to ensure notification were received in all states and setting up firebase in my computer was challenging choosing correct setting Android /firebase configuration was a challenge

Notification behaved as expected popup appeared in the app and they were stored, displayed in UI

CamScanner

Key steps I followed

* Creating app
* putting firebase to my PC using npm, doing firebase login setup firebase project and added android app in firebase console
* Added firebase depencies (firebase_core, firebase messaging) in the flutter project
* Configured Android for firebase update gradle files
* Initialized firebase and FCM in app main file
* Request notification permissions & handled device token retrieval
* Implement notification background foreground & terminated state
* Display received in UI & show popup dialog box
* Integrated Hive for persistent notifications storage & load history on app start
* Test by sending message from firebase console then verified display and delivery in the app

CamScanner